

Interoperability — built vs. yet to build

Standards-native by design. What's live in the prototype, and the honest roadmap to hospital- and practice-scale interoperability.

IN THE PROTOTYPE TODAY

- **FHIR R4 + US Core** data model; **USCDI v3** mapping.
- **SMART on FHIR** app surface — the (g)(10) sandbox.
- **Evidence / citation engine** (PMID · §170.315 · authority).
- **AI governance registry** (scale / pause / kill · PCCP).
- **Compliance engine** — 50-state + PR statutory deadlines.
- **Role UIs** — clinician, patient, researcher, CWO/CMO.

YET TO BUILD (PRIORITY ORDER)

- **Certified (g)(10) API** + Inferno conformance — productionize.
- **TEFCA / QHIN** join + **C-CDA** transitions of care.
- **Payer interop**: Da Vinci CRD/DTR/PAS (prior auth); X12 837/835.
- **Labs** (LOINC, HL7 v2 ↔ FHIR), **e-Rx** (NCPDP SCRIPT, EPCS), **imaging** (DICOMweb / FHIR ImagingStudy).
- **Vital records / EDRS** (VRDR FHIR) per jurisdiction.
- **Bulk data** (\$export / Flat FHIR) for registry, population & research.
- **Patient matching** + provider directory; **consent management**.
- **Documentation integrity**: signed notes + hash-chained, append-only audit log — the standards-based alternative to blockchain.
- **Low-resource / offline-tolerant sync** (LMIC / rural).
- **Security attestations**: SAFER, HIPAA, audit; SOC 2 / HITRUST.

PLUS: THE 5 READINESS FACTORS STILL TO CLOSE

FENNELLY · 15 FACTORS · PMID 33017724

Organizational & human (3)

Training — role-based + super-users · **Support model** — tiered go-live desk · **User competencies** — role-based proficiency, incl. nurses.

Technological (2)

Infrastructure — low-resource / offline-tolerant mode (LMIC / rural) · **Testing** — phased field pilots with conformance & fidelity gates.

SCALE — SAME CORE, TWO FOOTPRINTS

Hospital scale

Enterprise integration engine, HL7 v2 / HIE bridges, SSO, HA & DR, bulk export for the warehouse.

Private-practice scale

Turnkey SaaS, plug-and-play certified connectors, lightweight onboarding — no integration team.

Principle — certification by composition: certified modules on a FHIR-native core; every capability cites its §170.315 criterion. Interoperability is a returned-time feature, not a project.